

## Md. Syamul Bashar

Assistant Professor, Mechanical Engineering Department  
Shahjalal University of Science and Technology, Bangladesh

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### Research Interests

1. Physical AI
2. Intelligent Design & Manufacturing
3. Control Theory & System Dynamics

### Academic Credentials

**BSc. in Mechanical Engineering with Distinction** (January 2016 – November 2019)

Shahjalal University of Science and Technology (SUST), Bangladesh (**Acceptance rate below 3%**)

**CGPA: 3.78/4.00 (Ranked 2nd in Class)**

**Thesis Title:** Numerical Investigation of Blood Flow through Human Common Carotid Artery.

**Supervisor:** Md. Shafiqul Islam, Assistant Professor, Department of Mechanical Engineering, SUST

### Conference Publications

1. Haque, M. A., Shamim, F. M., & **Bashar, M. S.\* (2025)**. Sensor-Lite Model for Thermal Comfort Prediction Using Contextual Proxy Features, *15th International Conference on Mechanical Engineering (ICME 2025)*, Dhaka, Bangladesh, [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=6204578](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6204578).
2. Haque, M. A., Asaduzzaman, M., & **Bashar, M. S.\* (2025)**. Predicting Globe Temperature: Model Development and Analysis, *15th International Conference on Mechanical Engineering (ICME 2025)*, Dhaka, Bangladesh, [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=6200418](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6200418).
3. Haque, M. A., Shamim, F. M., & **Bashar, M. S.\* (2026)**. Reinforcement Learning for Thermal Comfort Control Using Predicted Mean Vote Setpoints, *IEEE International Conference on Engineering and Frontier Technologies 2026 (ICEFrontT 2026)*, Tangail, Bangladesh (Submitted).
4. Samin, A. A., Zihan, S. W., & **Bashar, M. S.\* (2026)**. Physics-Guided Surrogate Modeling for Airfoils' Aerodynamic Performance in Incompressible Flow Regime, *IEEE International Conference on Engineering and Frontier Technologies 2026 (ICEFrontT 2026)*, Tangail, Bangladesh (Submitted).
5. Samin, A. A. & **Bashar, M. S.\* (2026)**. Conditional Generative Inverse Design of Airfoils in the Incompressible Flow Regime, *International Conference on Emerging Frontiers in Advanced Sciences and Technologies 2026 (EFAST 2026)*, Pabna, Bangladesh (Submitted).
6. Raz, A.B., Haque, M. A., **Bashar, M. S.\* (2026)**. Goal-Aware Reinforcement Learning Policy for Lifelong Skill Acquisition in Robotic Manipulation, *International Conference on Emerging Frontiers in Advanced Sciences and Technologies 2026 (EFAST 2026)*, Pabna, Bangladesh (Submitted).

## Journal Publications

1. **Bashar, M. S.\***, Hossain, R., Rahman, M. H., Roy, U., & Islam, M. S. (2024). CFD analysis of non-Newtonian blood flow through human carotid artery bifurcation: Carotid sinus susceptible to atherosclerosis. *Heliyon*, November 08, 2024, <https://doi.org/10.1016/j.heliyon.2024.e40286>.
2. **Bashar, M. S.\***, Hossain, R., & Islam, M. S. (2022). CFD analysis of blood flow through bifurcated carotid artery (of human): Newtonian vs. non-Newtonian blood model. *SUST Journal of Science & Technology*, 32(2), December 31, 2022, <https://journals.sust.edu/index.php/sustjst/article/view/27>.
3. Haque, M. A., Shamim, F. M., & **Bashar, M. S.\***. Proxy Features-based Thermal Comfort Prediction with Minimal Sensor Requirements (*Manuscript ready; Extended version of a published conference article*).
4. Haque, M. A., Shamim, F. M., & **Bashar, M. S.\***. Adaptive Thermal Comfort Control Using Reinforcement Learning (*Manuscript ready; Extended version of an accepted conference article*).

## Undergraduate Thesis (Mentoring)

1. Samin, A. A. & **Bashar, M. S.\*** (2026). Conditional Airfoil Synthesis Using Generative Models in the Incompressible Flow Regime (*Ongoing*).
2. Raz, A.B. & **Bashar, M. S.\*** (2026). Adaptive Vision-Language-Action Policies for Variable-DoF Robot Manipulation: Benchmarking SmoVLA,  $\pi 0.5$ , and ACT (*Ongoing*).
3. Rahman, M.O., Hasan, M., & **Bashar, M. S.\*** (2026). Interpretable and Physics-Guided Reduced-Order Modeling for Stepped-Airfoil Aerodynamics (*Ongoing*).
4. Haque, M. A., Shamim, F. M., & **Bashar, M. S.\*** (2025). Reinforcement Learning for Thermal Comfort Control Using Predicted Mean Vote Setpoints.
5. Samin, A. A., Zihan, S. W., Haque, M. A., & **Bashar, M. S.\*** (2025). A Dataset & Data-Driven Surrogate Modeling for Airfoils' Aerodynamic Performance in Incompressible Flow Regime.
6. Ahamed, A., Khan, A. M., & **Bashar, M. S.\*** (2023). Performance Analysis of Various Simultaneous Localization and Mapping Algorithms using 2D Lidar with Robot Operating System.

## Teaching Experience (Undergraduate Level)

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|----------------------------|-----------------------|-------------------------------------|
| 1. Programming Methodology | 3. Numerical Analysis | 5. Fluid Mechanics                  |
| 2. Control Engineering     | 4. Applied Statistics | 6. Instrumentation and measurements |

## Project Grants

1. **Integrating Machine Learning with IoT for Smart Indoor Farming Management.**  
*Fund:* Approximately 3,000 USD in local currency | Funded by SUST | Funded by SUST (2023).  
*Objectives:* Trained the Yolo v11 model to detect lettuce plant diseases.
2. **Development of an Autonomous Boat for Floating Solid Waste Detection and Collection from Rivers.**  
*Fund:* Approximately 3,000 USD in local currency | Funded by SUST (2022).  
*Objectives:* Trained the Yolo v11 model to detect floating solid wastes.

## Skills and Language

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|---------------------------------------------------------|-------------------------|-------------------------------|
| 1. IELTS ( <i>Score 7.5; L: 8; R: 8; W: 7; S: 6.5</i> ) | 4. Python, C, MATLAB    | 7. Ansys Fluent               |
| 2. Deep Learning, Reinforcement Learning                | 5. ROS 2                | 8. Cross-platform Application |
| 3. Pytorch, Tensorflow                                  | 6. AutoCAD, Solid Works | Development                   |

## Awards and Recognition

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1. <b>Teaching Assistant</b> ( <i>July 2017 – December 2017</i> )<br><i>Course Title:</i> Programming Methodology<br><i>Course Teacher:</i> Professor Md. Ferdous Alam | 2. Shahjalal University of Science and Technology Merit Scholarship 2016, 2017, 2018, and 2019 |
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